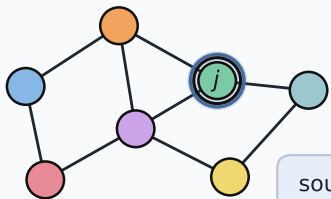


A Semantic intervention

Change X only; keep S fixed.

Clean graph



Donor content swap

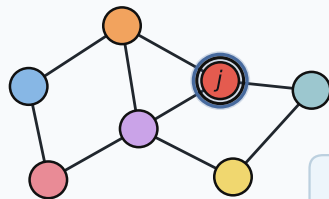
$x_j \leftarrow \tilde{x}$, with S fixed

donor row



average over K donors

Intervened graph



Visual key

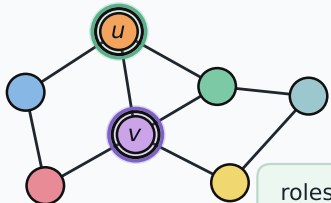
orange circle: inner fill: content row x_i

green halo: structural role / support s_i

B Structural intervention

Change S only; keep X fixed.

Clean graph



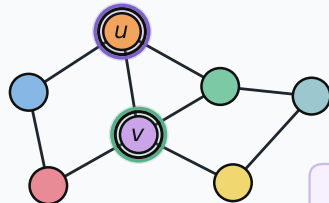
Structural transposition

$S' = P_{(uv)} S P_{(uv)}^\top$, with X fixed



average over K degree-matched partners

Intervened graph



Both rows then feed the same carriage estimator:
 $\Delta h_i = h_i^L(\text{clean}) - h_i^L(\text{intervention})$